

mock-data-generation

Mock Data Generation Prompt Chain

Use Case

Use this chain when a human analyst needs Edge Copilot to generate schema-valid synthetic data artifacts from compressed Oracle schema context.

Required Local Preparation

- Prepare compressed schema context with tables, columns, data types, PKs, FKs, nullability, and uniqueness notes where available.
- Decide target row counts, realism rules, date ranges, and domain constraints.
- Choose whether the final artifact should be SQL inserts, a PL/SQL block, a Python generator, or CSV files.
- Keep a way to inspect FK integrity and row counts locally before use.

Prompt Sequence

1. 01-schema-submission.md: submit compressed schema and generation objective.
2. 02-pk-fk-hierarchy-analysis.md: derive parent-child load order.
3. 03-output-mode-selection.md: choose the artifact style and assumptions.
4. 04-data-generation.md: generate the requested mock-data artifact.
5. 05-integrity-verification.md: review FK validity, constraints, and realism.

Related Procedure Closet

- `procedures/mock_data_generator/SKILL.md`

Related Reference Drawers

- `references/oracle-sql/sections/plsql_basics.md`
- `references/oracle-sql/sections/oracle_idioms.md`

Mock Data Generation Chain - 01 Schema Submission

Follow the MDCS procedure closet at `procedures/mock_data_generator/SKILL.md`.

Act as a Test Data Engineer generating schema-valid synthetic data. This is the first step of a multi-turn mock data workflow.

Compressed schema context:

<paste compressed schema context here>

Generation objective:

<paste target tables, row counts, realism rules, date ranges, domain rules, and preferred output format>

Return:

1. Restated mock data objective.
2. Missing schema details needed for constraint-safe generation.
3. Questions about row counts, realism, nullability, uniqueness, and output format.
4. Whether Oracle-specific drawer guidance may be needed later.

Do not generate data yet.

Mock Data Generation Chain - 02 PK/FK Hierarchy Analysis

Continue following `procedures/mock_data_generator/SKILL.md`.

Using the compressed schema already provided, derive the parent-child dependency order before generating any data.

Return:

1. Root parent tables.
2. Child tables and FK dependencies.
3. Proposed generation/load order.
4. Key pools that must be reserved for child rows.
5. Constraint details that remain ambiguous.

Keep this as a planning step only. Do not generate the final artifact yet.

Mock Data Generation Chain - 03 Output Mode Selection

Continue following `procedures/mock_data_generator/SKILL.md`.

Select the output mode for the mock data artifact based on the schema, dependency plan, and my local loading workflow.

Preferred output mode:

<paste one of: SQL INSERT statements, PL/SQL block, Python generator, CSV files, or undecided>

Local loading constraints:

<paste transaction, sequence, file, delimiter, environment, or review constraints here>

Return:

1. Recommended output mode.
2. Why it fits the dependency and loading constraints.
3. Oracle reference drawer to consult, if PL/SQL or Oracle idioms are needed.
4. Assumptions to confirm before final generation.

Mock Data Generation Chain - 04 Data Generation

Continue following `procedures/mock_data_generator/SKILL.md`.

Generate the mock data artifact now using the confirmed schema, dependency order, row counts, realism rules, and output mode.

Confirmed output mode:

<paste confirmed output mode here>

Final generation constraints:

<paste final row counts, fixed values, date ranges, domain values, nullability, uniqueness, >

Return exactly one artifact style. Include:

1. The generated artifact content.
2. Table generation order.
3. Row-count summary.
4. Assumptions and unresolved gaps.
5. Validation checks I should run locally.

Use `references/oracle-sql/sections/plsql_basics.md` or `references/oracle-sql/sections/oracle_id` only when the chosen artifact requires those Oracle details.

Mock Data Generation Chain - 05 Integrity Verification

Continue following `procedures/mock_data_generator/SKILL.md`.

Review the generated mock data artifact and my local verification results.

Artifact or summary:

<paste generated artifact, table summaries, or relevant excerpts here>

Verification results:

<paste FK checks, row counts, uniqueness checks, nullability checks, date/domain realism not

Return:

1. FK integrity assessment.
2. Required-field and uniqueness assessment.
3. Row-count and load-order assessment.
4. Realism assessment against stated rules.
5. Concrete fixes needed before use.